BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 236 · Writing, following and documenting algorithms

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Give a test for a range 0–100 that should be rejected.

Q2. What does return do?

Q3. What is the last index for an array of length 4?

Q4. Why is a[0] unsafe for an empty array?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

algorithm — Precise finite steps that solve a problem.

precondition — Requirement inputs must meet before execution.

postcondition — Condition guaranteed if the algorithm completes correctly.

invariant — Property maintained during repeated steps.

documentation — Clear description of purpose, inputs and expected behaviour.

New learning

State the precondition — The array must not be empty, or a[0] cannot be read.

Trace the maximum — Set best to 3; compare 7, update best to 7, then ignore 2 and 5.

Document contract — Input: nonempty int array. Output: greatest element. Does not modify input.

Worked example

Trace Largest([4,2,9]) and write the returned result.

1. Predict first — Start best at first element 4; 2 does not replace it.

2. Model and explain — 9 is larger so best becomes 9; return 9.

We Do · Practise with guidance

Trace Largest([8,11,6,10]) and state its precondition.

```
best=8; process 11, 6, 10;
```

Predict first

11 replaces initial best 8; neither 6 nor 10 exceeds 11.

Model and explain

The output is 11; the required precondition is a nonempty array.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Trace Largest([2,5,1]) and state initial and final best.

Question 2

Write pseudocode to return the minimum of a nonempty array values and explain its precondition.

Question 3

Evaluate what happens for Largest([-4,-2,-9]) and explain why initialising best=0 would be incorrect.

Exit Ticket · Answer before checking

Initialising a maximum-tracking variable to 0 is correct for every integer array.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. −1 or 101.

Q2. Passes a function's result back to the caller.

Q3. 3.

Q4. Index 0 does not exist, causing an out-of-range error.

You Do pages · Independent questions

Q1. [4] Starts best=2 [1]; 5 greater so best=5 [1]; 1 ignored [1]; returns 5 [1].

Q2. [6] Set smallest=values[0] [1]; loop over remaining items [1]; if item<smallest [1] then smallest=item [1]; return smallest [1]; array must be nonempty [1].

Q3. [6] Original best=−4 [1]; update to −2 [1]; ignores −9 [1]; returns −2 [1]; initialising to zero would incorrectly return 0 [1] even though zero is absent [1].

Exit Ticket · Model answer

False. Arrays containing only negative values would produce an incorrect result.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: